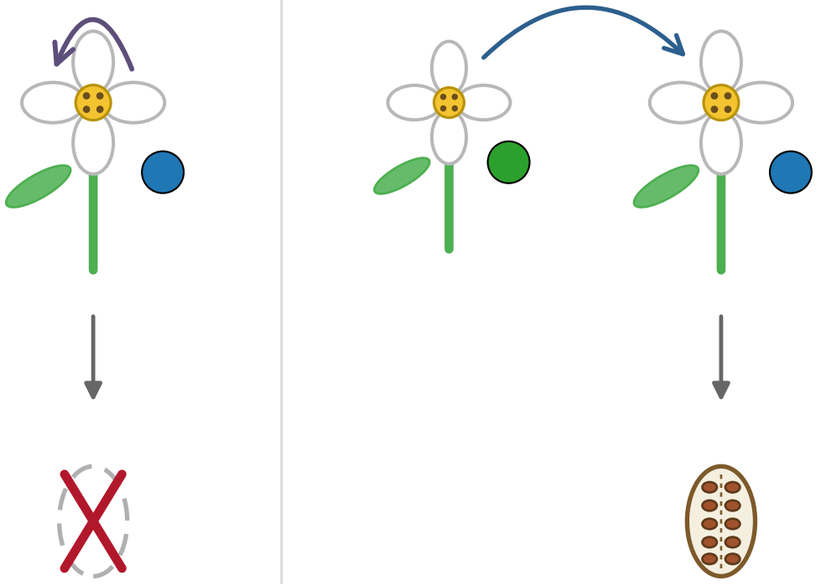
1. Plants reject their own pollen

colour = S-type
(32 known in LEPA)

self-incompatible

own pollen

other S-type



no seed

seed

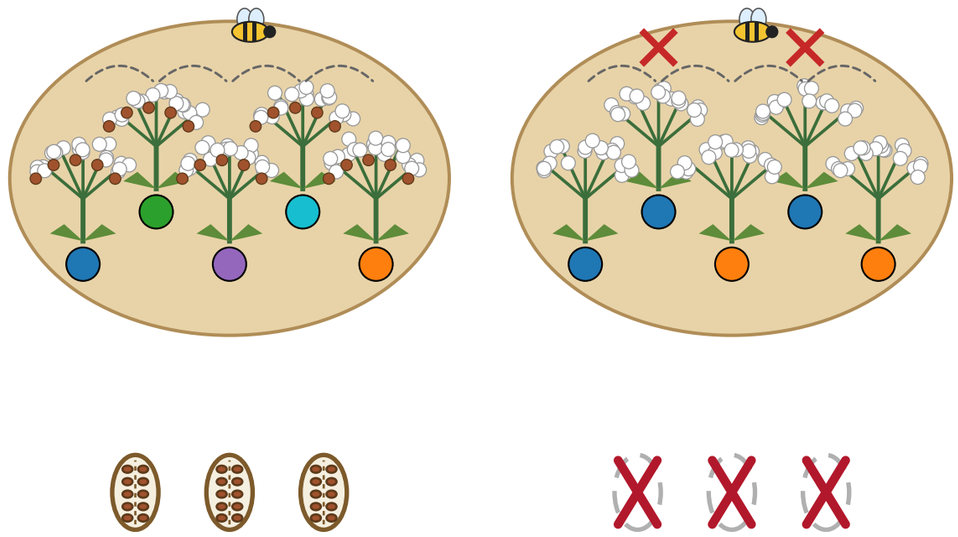
the S-locus (a male + a female gene) gives each plant an S-type: pollen with the flower's own S-type is rejected

2. Few S-types → mate limitation → population decline

mate limitation

many S-types

few S-types

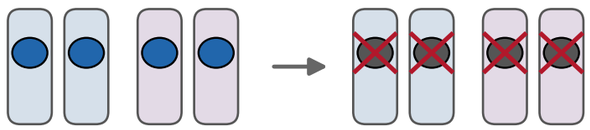


in the field: less seed than plant size predicts

small, isolated populations lose S-types:
plants run out of compatible mates and set little seed

3. Some plants broke the gene

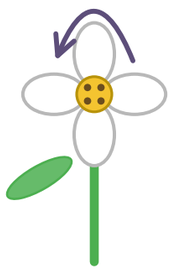
self-compatible



working S-locus
(4 copies)

broken copies

own pollen



LEPA — 263 plants genotyped

247 self-incompatible

15 partly broken

1 self-compatible

in the field: more seed than plant size predicts



seed

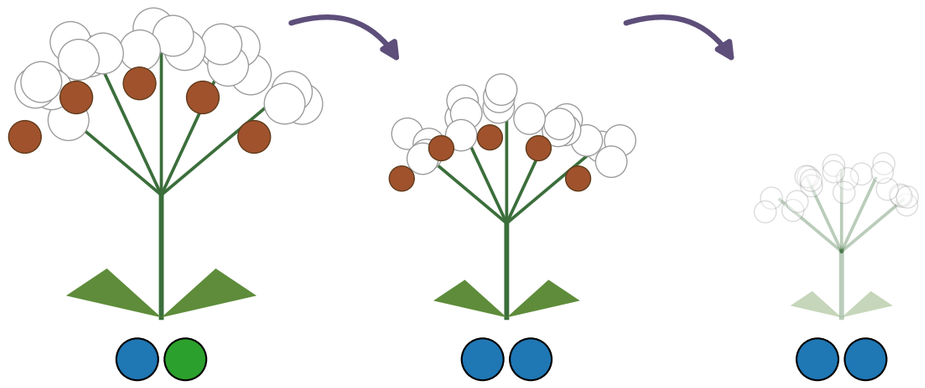
plants with broken S-locus copies accept their own pollen and set seed alone

4. Selfing → inbreeding → reduced fitness → population decline

inbreeding

self

self



parent

selfed
offspring

next
generation

seed ↓

survival ↓

diversity ↓

selfed offspring grow more alike and weaker, generation after generation